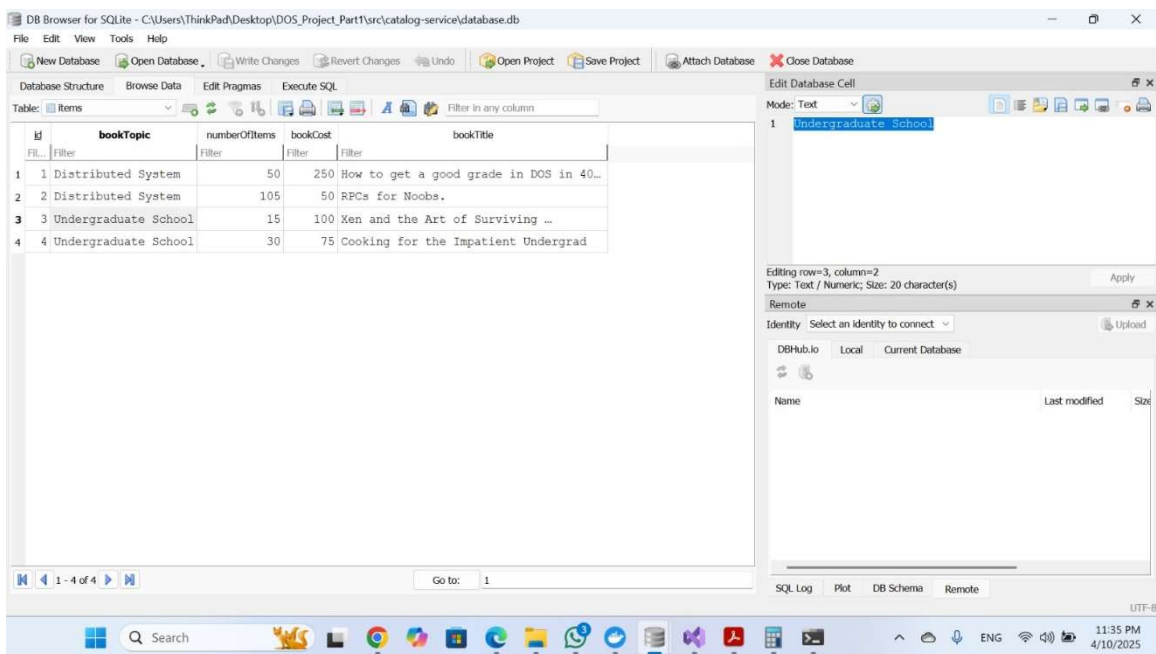


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Distributed Operation Systems – 10636456
Project – Part 1

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In this project we design a smallest book store , in this project we have only 4 books , the data base is include the books each book have (ID , Book topic , Number of item , Book Title) .



We First create a Main folder, the main folder contain a **Dockerfile** to run the containers , **Docker-Compose** to run the multi-containers , two packages for JSON , **Node-module** (it create automatically) , Finally **src folder** , in this folder we have 3 micre-services , one of thim for Front-end his name is **Client-Service** , and two for Back-end , there name **Ordder-service** , **Catalog-Service** , and one folder for **nginx** this file to communication between services .

To run our project we use this command :

```
cd C:\Users\ThinkPad\Desktop\DOS_Project_Part1 .
```

```
docker-compose up -d --build .
```

```
cd C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service .
```

```
node index.mjs sb or node index.mjs ib , node index.mjs pb.
```

The result of our project : (we worked in 3 operation) :

1- Search about Book in data base using book topic :

```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>node index.mjs sb
? Enter the topic of the book to view its details : Undergraduate School
Response Data: [
  {
    id: 3,
    bookTopic: 'Undergraduate School',
    numberOfItems: 5,
    bookCost: 1000,
    bookTitle: 'Xen and the Art of Surviving Undergraduate School'
  },
  {
    id: 4,
    bookTopic: 'Undergraduate School',
    numberOfItems: 13,
    bookCost: 900,
    bookTitle: 'Cooking for the Impatient Undergrad'
  }
]
```

```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>
```

2- Display the information about book using item number :

```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>node index.mjs ib
? Enter the item number of the book to get information about it: 1
Response Data: { item: [ { id: 1, numberOfItems: 50, bookCost: 250 } ] }
```

```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>node index.mjs ib
? Enter the item number of the book to get information about it: 2
Response Data: { item: [ { id: 2, numberOfItems: 105, bookCost: 50 } ] }
```

```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>node index.mjs ib
? Enter the item number of the book to get information about it: 3
Response Data: { id: 3, numberOfItems: 15, bookCost: 100 }
```

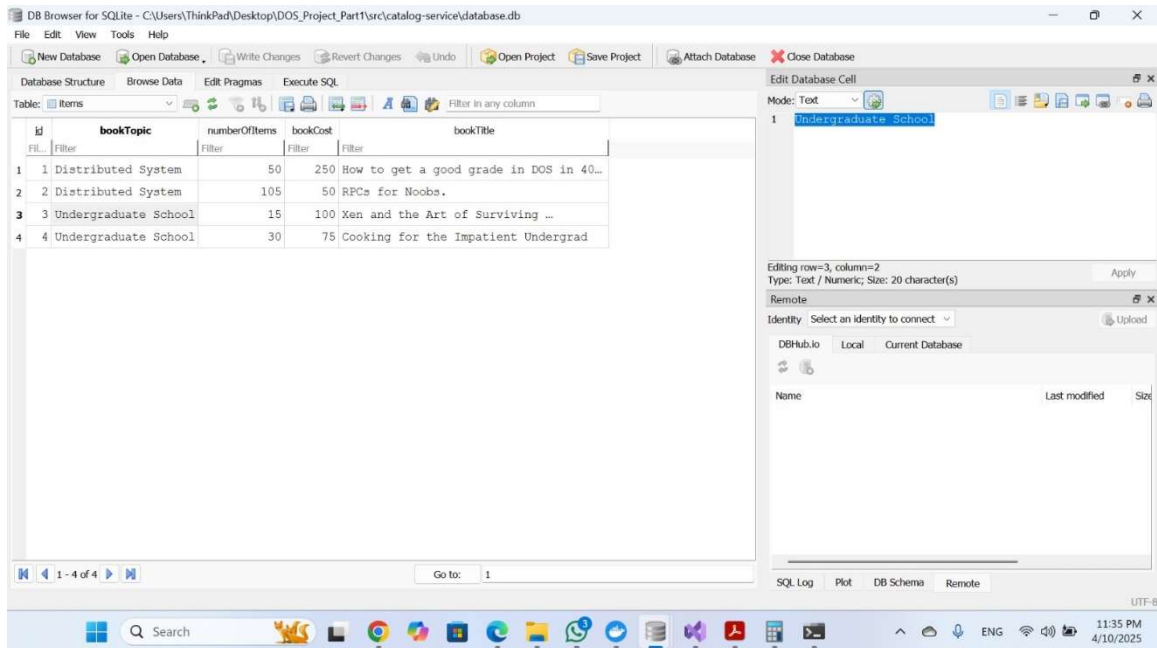
```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>node index.mjs ib
? Enter the item number of the book to get information about it: 4
Response Data: { item: [ { id: 4, numberOfItems: 30, bookCost: 75 } ] }
```

```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>
```

3- Purchase a book using item number :

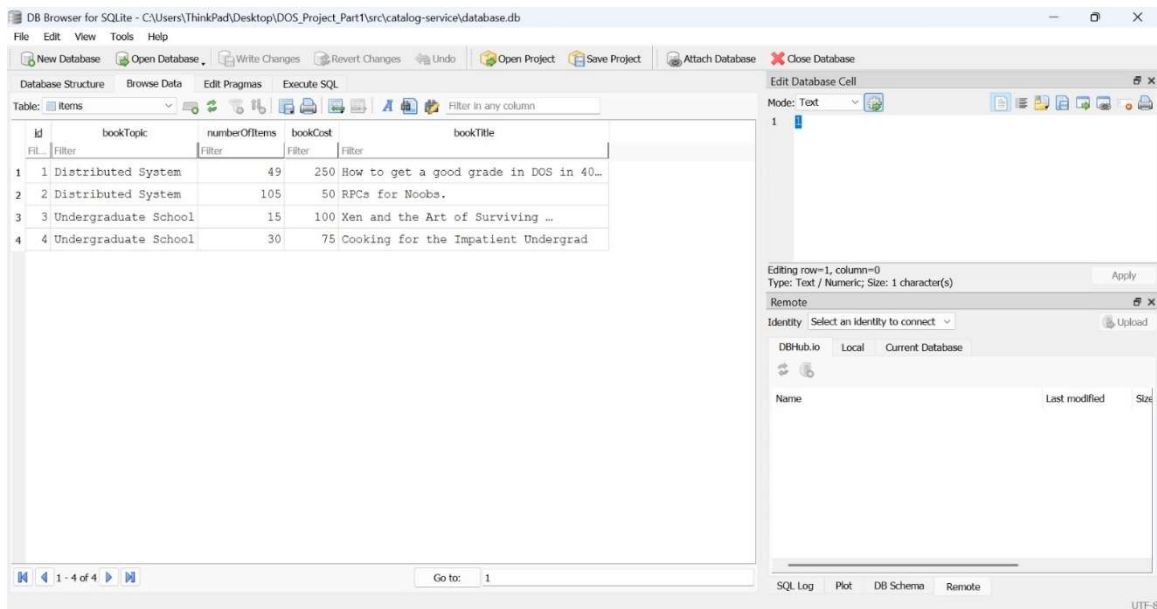
```
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>node index.mjs pb
? Enter the item number of the book to purchase it: 1
? Please enter the payment amount: 2000
Response Data: { message: 'The request has been sent successfully ^_^' }
C:\Users\ThinkPad\Desktop\DOS_Project_Part1\src\client-service>
```

The data base befor Purchase operation



id	bookTopic	numberOfItems	bookCost	bookTitle
1	Distributed System	50	250	How to get a good grade in DOS in 40...
2	Distributed System	105	50	RPCs for Noobs.
3	Undergraduate School	15	100	Xen and the Art of Surviving ...
4	Undergraduate School	30	75	Cooking for the Impatient Undergrad

The data base after Purchase operation , the number of item has ID1 is decreased by 1.



id	bookTopic	numberOfItems	bookCost	bookTitle
1	Distributed System	49	250	How to get a good grade in DOS in 40...
2	Distributed System	105	50	RPCs for Noobs.
3	Undergraduate School	15	100	Xen and the Art of Surviving ...
4	Undergraduate School	30	75	Cooking for the Impatient Undergrad

We used Node.js, Docker Desktop, SQLite as the database , and guthub to complete our project.